FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE JEE (Advanced), 2015

FULL TEST – I

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 240**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 04)** contains 4 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (05 – 08) contains 4 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

   Section-A (09 to 11) contains 1 paragraph having 3 questions. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct matching. No marks will be given for any wrong matching in any question. There is no negative marking.

3. **Section-C (01 – 06)** contains 6 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

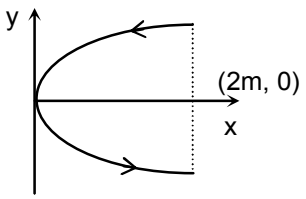
PART – I

SECTION – A

(Only One Option Correct Type)

This section contains **4 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

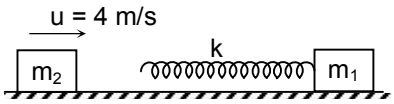
1. A conducting wire is bent in the form of a parabola $y^2 = 2x$ carrying a current $i = 2A$ as shown in the figure. The wire is placed in a uniform magnetic field $\vec{B} = -4\hat{K}$. The magnetic force on the wire is



(A) $-16\hat{i}$ (B) $32\hat{i}$
(C) $-32\hat{i}$ (D) $16\hat{i}$

2. If M = mass, L = length, T = time and I = electric current, then the dimensional formula of resistance R will be given by
(A) $[R] = [M^1L^2T^{-3}I^{-2}]$ (B) $[R] = [M^1L^2T^{-3}I^2]$
(C) $[R] = [M^1L^2T^3I^{-2}]$ (D) $[R] = [M^1L^2T^3I^2]$

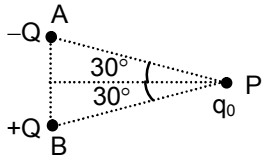
3. A block of mass $m_1 = 3$ kg is connected with ideal spring of spring constant k = 30000 N/m and kept at horizontal frictionless surface as shown in the figure. The block $m_2 = 1$ kg is moving with velocity u = 4 m/s towards block m_1. v_1 and v_2 are the final speed of the block m_1 and m_2 respectively.



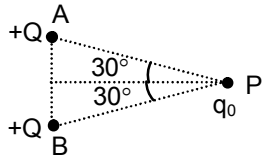
Choose the correct statement(s)
(A) $v_1 = v_2 = 1$m/s
(B) The maximum compression in the spring is 2 cm
(C) The maximum compression in the spring is 4 cm
(D) none

Space for Rough work

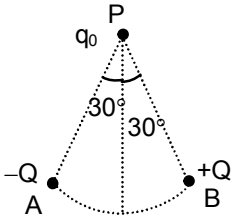
4 The diagram shows four systems of charged particles. Each system contains two charged particles at point A and point B as shown in the figure. A third positive point charge q_0 is kept at point P in each system. The direction of electrostatic force experienced by the charge q_0 will be



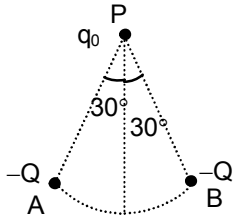
System -1; AP = BP = r and AB is a straight line



System -2; AP = BP = r and AB is a straight line



System -3; AP = BP = r and AB is a circular arc of radius r and centre at O



System -4; AP = BP = r and AB is a circular arc of radius r and centre at O

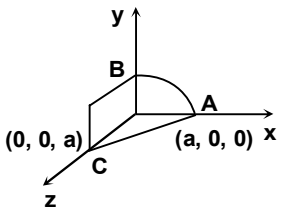
	System -1	System-2	System-3	System-4
(A)	↑	→	←	↓
(B)	↓	←	→	↑
(C)	↑	→	→	↓
(D)	↑	→	←	↑

Space for Rough work

(One or More than One Options Correct Type)

This section contains **4 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

5. In the given figure a wire loop has been bent so that it has three segments. Segment AB is a quater circle in xy plane, segment BC is L–shaped wire in yz-plane, and segment CA is straight line in xz plane. Here are three choices for a magnetic field through the loop.



(1) $\vec{B}_1 = 3\hat{i} + 7\hat{j} - 5t\hat{k}$

(2) $\vec{B}_2 = 5t\hat{i} - 4\hat{j} - 15\hat{k}$

(3) $\vec{B}_3 = 2\hat{i} - 5t\hat{j} - 12\hat{k}$

If the induced current in the loop due to $\vec{B}_1, \vec{B}_2$ and $\vec{B}_3$ are i_1, i_2 and i_3 respectively then:

(A) $i_1 < i_2$ (B) $i_1 > i_3$

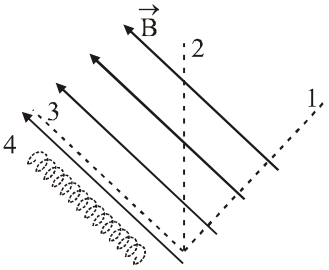
(C) $i_3 > i_2 > i_1$ (D) $i_1 = i_2 = i_3$

6. The surface tension phenomenon is the result of the tendency of a system to keep total

(A) density of the system is minimum (B) potential energy minimum

(C) volume of the system is minimum (D) surface area of the system is minimum

7. Radiation coming from a radioactive substance is passed through a cloud-chamber in which strong constant magnetic field is switched on in the direction shown in figure. Figure shows path followed by few particles coming from radioactive substance (shown by dotted lines) Neglect the interaction between different particles. Tick the correct statements :



(A) Path 1 can be followed by γ-radiation only

(B) Path 3 can be followed by γ-radiation only

(C) Path 4 can be followed by α-radiation only

(D) Path 4 can be followed by β-radiation

Space for Rough work

8. A particle of mass 'm' starts from origin and move in x-y plane velocity of particle is given by

$$\vec{v} = k_1 i + k_2\sqrt{1-\alpha y}\,\hat{j}$$

where k_1, k_2 & α are constant and 'y' is y-coordinate of position of particle. Choose the correct statements :

(A) Average force acting on particle $\frac{m\alpha k_2^2}{2}$

(B) Force acting on particle decreases first then increases

(C) Path of particle will be hyperbolic.

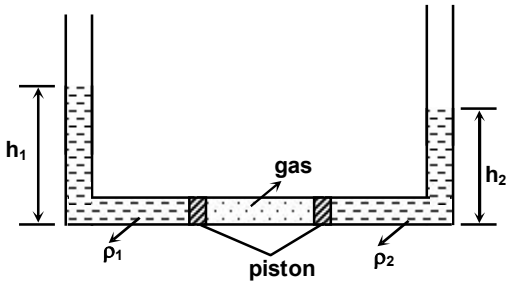
(D) Radius of curvature of the path followed by particle will decrease first then increase.

Comprehension type (Only One Option Correct)

This section contains **1 paragraph**, describing theory, experiments, data etc. Three questions relate to the one paragraph. Each question has **only one** correct answer among the four given options (A), (B), (C) and (D).

Paragraph for Questions 09 & 11

Figure shows a tube whose two limbs are vertical and one is horizontal of sufficient length. An ideal gas is enclosed between two massless non conducting movable piston of cross-sectional area 10 cm^2. A liquid of density ρ_1 = 1000 kg/m^3 and ρ_2 = 2000 kg/m^3 are poured in the vertical limbs of height of h_1 = 2m and h_2. Now gas is slowly heated. Consider the wall of tube is insulated.



9. Ratio of h_1 and h_2 at any instant will be

(A) $\frac{1}{2}$ (B) $\frac{2}{1}$

(C) $\frac{3}{2}$ (D) not constant

10. If due to heating separation between piston is increased by 3m then displacement of piston on left will be

(A) 2m (B) 1m

(C) 2.5 m (D) 1.5 m

11. Find the work done by gas

(A) 70 Joule (B) 390 Joule

(C) 90 Joule (D) 350 Joule

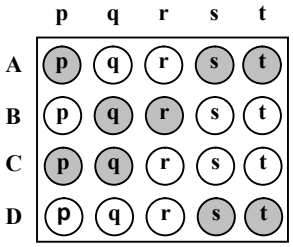
Space for Rough work

SECTION - B

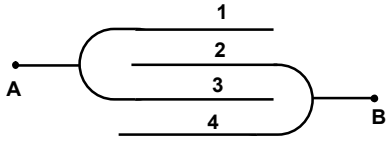
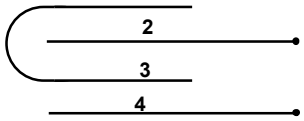
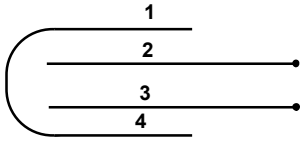
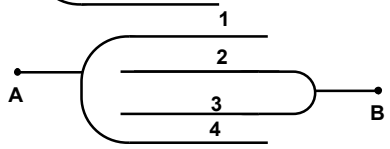
Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

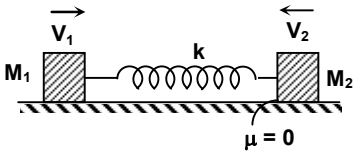
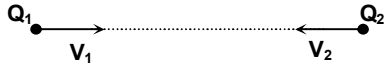
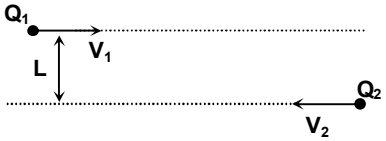
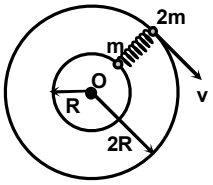


1. Terminal A is connected to positive terminal of battery and B is connected to negative terminal of plate and it send same charge Q in all four combination. Each plate have same area and separation between two consecutive plate is same.

Column I		Column II	
(A)		(p)	Top surface of plate 1 have zero charge
(B)		(q)	Upper surface of plate 4 have –Q charge
(C)		(r)	Same charge Appear on both surface of plate 2
(D)		(s)	Bottom surface of plate 4 have zero charge
		(t)	Total charge on plate 2 is Q

Space for Rough work

2. Column I shows four physical situation and relates these with column II.

Column I		**Column II**	
(A)	 Blocks are given velocities as shown on smooth horizontal surface	(p)	Mechanical energy is conserved
(B)	 Two positive charge particles are given velocities towards each other in a gravity free space as shown.	(q)	Momentum is conserved
(C)	 Two positive charge particles are given velocities along different line towards each other in a gravity free space as shown.	(r)	K.E. of centre of mass will remain constant
(D)	 Beads are free to move on a smooth circular rings placed on a smooth horizontal table.	(s)	At maximum or minimum separation K.E. in centre of mass frame will be zero or minimum.
		(t)	Angular momentum is conserved about any point in inertial frame

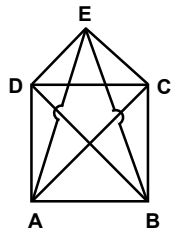
Space for Rough work

SECTION – C

(One Integer Value Correct Type)

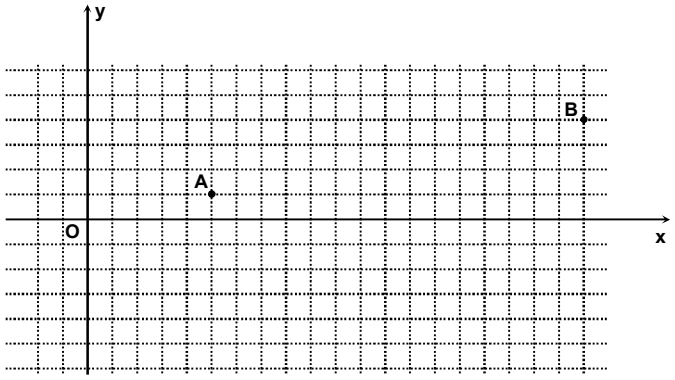
This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. In the given circuit the resistance of each sides and diagonals is $R\left(=\frac{35}{2}\Omega\right)$. The equivalent resistance across the points A and B is (given ABCD is a square)



2. If the sound heard by observer, whose equation is given as
$y = 8\sin 10\pi t \cos 200\pi t$ at $x = 0$
The number of beat frequency heard by observer is 2k, then the value of k is

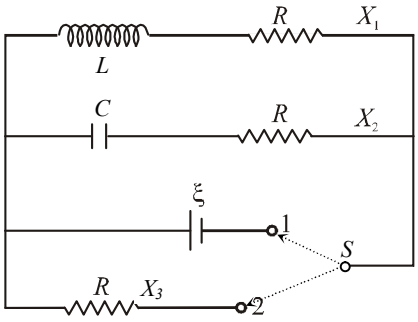
3. Point B is the image of point A due to reflection from mirror (which is either spherical concave mirror, or spherical convex mirror or plane mirror), whose principal axis is x-axis.



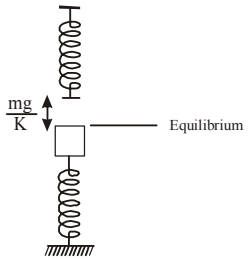
One square in x-axis represents 5 cm and one square on y-axis represents 1mm. The co-ordinate of centre of curvature of mirror is $(x_1, 0)$ and co ordinate of pole of mirror is $(x_2, 0)$ respectively. If $|x_1| + |x_2| = 5K$ cm, then find the value of K.

Space for Rough work

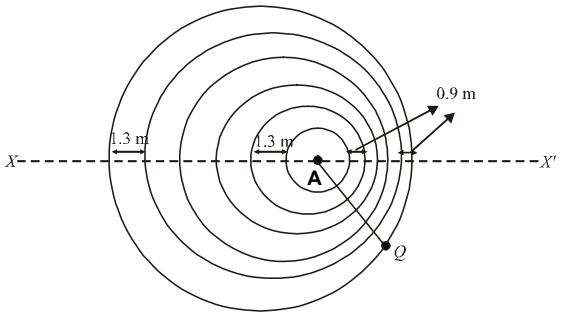
4. Figure shows a circuit containing three resistor X_1, X_2 & X_3 having resistance R each, an inductor, capacitor and an emf source having inductance 'L', capacitance C & emf ξ respectively (Given $R = \sqrt{\frac{L}{C}}$). The switch is first connected to position-1. When charge on capacitor becomes half of it maximum possible value then switch S is connected to position-2. Current in resistance X_3 just after shifting the switch from position-1 to position-2, is
(given L = 5mH, C = 2μF, R = 10 Ω, ξ = 5V)



5. In the figure shown a block of mass m is attached in a light spring of spring constant K and an identical spring hangs from ceiling. Initially lower spring is compressed in a state with compression equal to $\frac{3mg}{K}$ from natural length of spring when block is released it strikes upper spring and sticks to it. Amplitude of oscillation (in cm) is given mg = 10 N and $K = 100\sqrt{7}$ N/m



6. Figure shows wave fronts coming form a source at equal interval of $\frac{10}{3}$ milli second. Frequency of wave emitted by source is 300 Hz. XX' is line along common diameter of wave fronts passing through 'A'. Distance between two consecutive wave fronts along line XX' is 0.9 m and 1.3 m in the right and left of point A respectively. At some instant source is at point A. Detector is placed at point 'Q'. AQ makes an angle 60° with line XX' If frequency of wave received by detector is f Hz, then $\frac{f}{110}$ (in Hz) is equal to



Space for Rough work

Chemistry

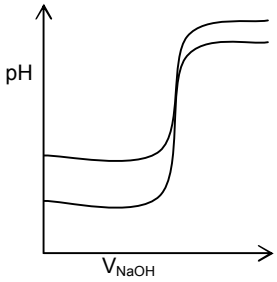
PART – II

SECTION – A

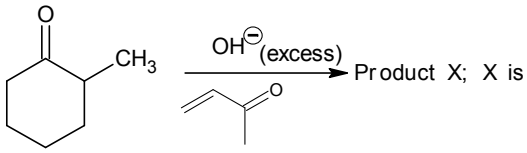
(Only One Option Correct Type)

This section contains **4 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

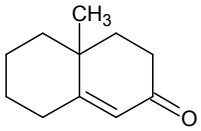
1. True or False
   (a) In any strong acids solution, the concentration of $[OH^-]$ will be zero.
   (b) If ΔG° of a reaction is positive, then the reaction will not proceed at all, in the forward direction for any concentration of reactants and products.
   (c) When titration curves are drawn for
   (i) 1 M HCl (50 mL) with 1 M NaOH and
   (ii) 0.01 M HCl (50 mL) with 0.01 M NaOH on the same graph paper they look like

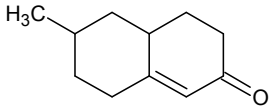


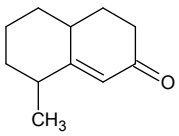
(A) TTF
(B) FFT
(C) FTF
(D) TFT

2. 

Product X; X is

(A) 

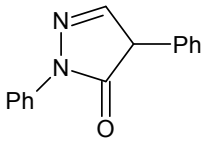
(B) 

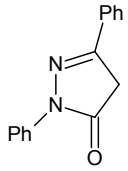
(C) 

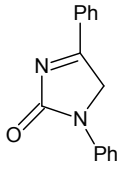
(D) None of these

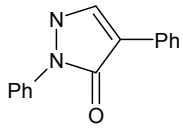
Space for Rough work

3. 

(A) 

(B) 

(C) 

(D) 

4. Which of the following statement is false?
(A) In $XeOF_4$, the central atom Xe doesn't lie on the plane of the square
(B) In PF_2Cl_3, both the F-atoms lie on the axial bond
(C) In PCl_5, the axial bond length is longer than the equatorial bond length
(D) The dipole moment of CH_3F is greater than CH_3Cl

Space for Rough work

(One or More than One Options Correct Type)

This section contains **4 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

5. Which of the following is correct for lyophilic aquasols?
   (A) Its surface tension is lower than that of H_2O
   (B) Its viscocity is higher than that of water
   (C) Colloidal silver
   (D) Colloidal antimony

6. Which of the following metals can be obtained by reducing its metal oxide using aluminium?
   (A) Ca (B) Fe
   (C) Mg (D) Cr

7. Select wrong statements
   (A) Phenol reacts with Na_2CO_3 and librates CO_2 gas
   (B) Phenol turn blue litmus to red
   (C) Reactivity of methanol with sodium metal is more than that of isopropyl alcohol
   (D) Methanol gives iodoform test

8. Borax bead test is given by
   (A) Co^{2+} (B) Zn^{2+}
   (C) Cu^{2+} (D) Ni^{2+}

Space for Rough work

Comprehension type (Only One Option Correct)

This section contains **1 paragraph**, describing theory, experiments, data etc. Three questions relate to the one paragraph. Each question has **only one** correct answer among the four given options (A), (B), (C) and (D).

Paragraph for Questions 09 & 11

Electrolysis is the process in which electrical energy is converted to chemical energy. In electrolytic cell, oxidation takes place at anode and reduction at cathode. Electrode process depends on the nature of electrode used for electrolysis. Amount of substance liberated at an electrode is directly proportional to the amount of charge passed through it. The mass of substance liberated at electrode is calculated by using the following relation:

$$m = \frac{ItE}{96500}$$

Here, E represents the equivalent mass and 96500 C is called the Faraday constant. Faraday (96500C) is the charge of 1 mole electron, i.e., 6.023×10^{23} electrons. This much charge is used to liberate one gram equivalent of any substance at each electrode.

9. The platinum electrodes were immersed in a solution of cupric sulphate ($CuSO_4$) and electric current is passed through the solution. After some time, it was observed that the colour of copper sulphate disappeared with evolution of a gas at the electrode. The colourless solution contains:
   (A) platinum sulphate (B) copper nitrate
   (C) copper sulphate (D) sulphuric acid

10. The passage of current liberates H_2 at cathode and Cl_2 at anode. The solution is:
   (A) copper chloride in water (B) NaCl in water
   (C) mercuric chloride in water (D) $AuCl_3$ in water

11. Calculate the volume of gas liberated at the anode at S.T.P. during the electrolysis of a $CuSO_4$ solution by a current of 1 A passed for 16 minutes and 5 seconds:
   (A) 224 mL (B) 56 mL
   (C) 112 mL (D) 448 mL

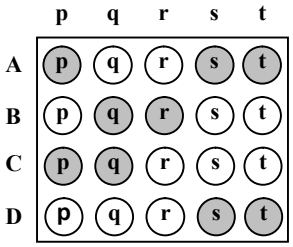
Space for Rough work

SECTION - B

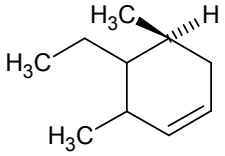
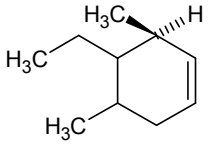
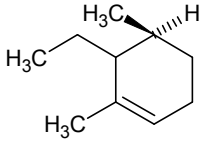
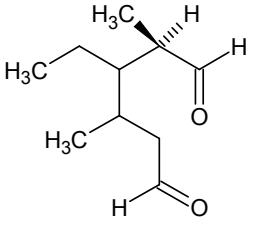
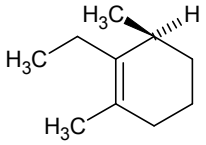
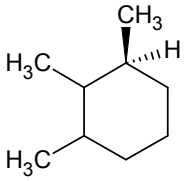
Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

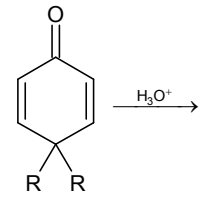
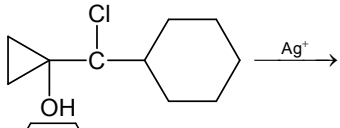
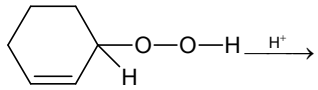
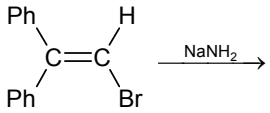


1. Match the following:

Column – I	Column – II
(A) When reacts with Br_2/CCl_4, diastereomer will be formed	(p) 
(B) When reacts with O_3/H_2O, dicarboxylic acid will be formed	(q) 
(C) Optically active compound	(r) 
(D) When undergo reductive ozonolysis following product will be formed 	(s) 
	(t) 

Space for Rough work

2. Match the following:

Column – I		Column – II	
(A)		(p)	Final major product has a six membered ring
(B)		(q)	One of the group will undergo migration
(C)		(r)	Aldol condensation is one of the step
(D)		(s)	Final compound has three degree of unsaturation
		(t)	Contain at least one π bond.

Space for Rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Two aqueous solutions are put in an evacuated chamber. When equilibrium is obtained, it is found that one solution contains 0.01% of CH_3COOH and other 0.014% of urea by weight, if degree of dissociation is 'α' then what is the value of 10α?

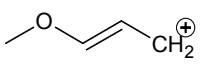
2. Metal element 'M' of radius 50 nm is crystallized in FCC format and make cubical crystal such that face of unit cells aligned with face of cubical crystal. If total number of metal atoms of 'M' at face of cubical crystal is 6×10^{30} then area of one face of cubical crystal is $A \times 10^{16}$ m^2, the value of A is

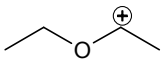
3. A 10 L box contains 41.4 gram of a mixture of gases C_xH_8 and C_xH_{12}. The total pressure of 44°C in flask is 1.56 atm. Analysis revealed that the gas mixture has 87% total C and 13% total H. Find out the value of x.

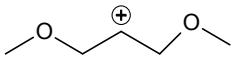
4. In the reaction $A + B \rightleftharpoons AB$, if the concentration of A and B is increased by a factor of 2, it will cause the equilibrium concentration of AB to change to……..

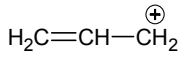
5. A solution of $SrCO_3$ is saturated with respect to SrF_2. Ksp = 7.9×10^{-10}. If the fluoride ion concentration is found to be 4.0×10^{-2}. Then the carbonate ion is found to be 1.4×10^{-x} M. Find the value of x.

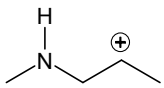
6. Which of the following carbocation have stability greater than an isopropyl cation $(CH_3)_2CH^+$

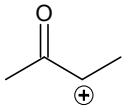

(i)

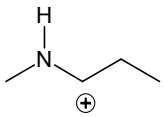

(ii)


(iii)


(iv)


(v)


(vi)


(vii)

Space for Rough work

Mathematics PART – III

SECTION – A

(Only One Option Correct Type)

This section contains **4 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

1. If f: R $\to$ R is a continuous function satisfying f(0) = 1 and f(2x) – f(x) = x $\forall$ x $\in$ R and $\lim_{x\to\infty}\left(f(x)-f\left(\frac{x}{2^n}\right)\right)=P(x)$ then P(x) is
   (A) a constant function (B) a linear polynomial in x
   (C) a quadratic polynomial in x (D) a cubic polynomial in x

2. If f(x) is continuous function such that f(x) > 0 $\forall$ x $\geq$ 0 and $\left(f(x)\right)^{2014}=1+\int_0^x f(t)dt$ then the value of $(f(2014)^{2013}$ is equal to
   (A) 2013 (B) 2014
   (C) 2015 (D) 2016

3. An exam consist of 3 problems selected randomly from a collection of 2n problems, where n is an integer greater then 1. For a student to pass, he needs to solve correctly at least two of the three problems. If the student knows to solve exactly half of 2n problems, then the probability that the student pass the exam is
   (A) $\frac{1}{2}$ (B) $\frac{1}{3}$
   (C) $\frac{3}{4}$ (D) cannot be determined

4. If the radius of in-circle and radii of ex-circles of a triangle are consecutive terms of a geometric progression then the largest angle of the triangle will be
   (A) 60º (B) 90º
   (C) 120º (D) none of these

Space for rough work

(One or More than One Options Correct Type)

This section contains **4 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

5. A point Q moves on the positive y-axis and a point P follows the point Q such that the motion of P is always towards Q and distance between P and Q is always equal to some constant K > 0. If initially Q is at origin and P is at (K, 0) then

(A) the differential equation representing the curve traced by P is $\frac{dy}{dx} = \frac{\sqrt{K^2 - x^2}}{-x}$

(B) y-axis is asymptote to the curve traced by P

(C) the equation of curve traced by P is $y = -K\ln\left(\frac{K + \sqrt{K^2 - x^2}}{x}\right) - \sqrt{K^2 - x^2}$

(D) the curve traced by P will has x-axis as asymptote

6. If a function y = f(x) is such that f′(x) is continuous function and satisfies

$(f(x))^2 = K + \int_0^x \left(\left(f(t)\right)^2 + \left(f'(t)\right)^2\right)dt$, $K \in R^+$, then

(A) f(x) is increasing function $\forall\ x \in R$ (B) f(x) is bounded function
(C) f(x) is neither even nor odd (D) if K = 100 then f(0) = 10

7. Consider parabola $P_1 \equiv y = x^2$ and $P_2 \equiv y^2 = -8x$ and the line $L \equiv \ell x + my + n = 0$. Which of the following holds true (a point (α, β) is called rational point if α and β are rational)

(A) if ℓ, m, n are odd integers then the line L can not intersect parabola P_1 in a rational point

(B) line L will be tangent to P_1 if m, $\frac{\ell}{2}$, n are in G.P.

(C) if line L is common tangent to P_1 and P_2 then $\ell + m + n = 0$

(D) if line L is common chord of P_1 and P_2 then $\ell - 2m + n = 0$

8. If H_n denotes the number of ways in which n horses participating in a race can reach the finishing line, when one or more horses can reach the finishing line at the same time, then

(A) $H_2 = 3$ (B) $H_3 = 13$
(C) $H_4 = 75$ (D) $H_3 = 10$

Space for rough work

Comprehension type (Only One Option Correct)

This section contains **1 paragraph**, describing theory, experiments, data etc. Three questions relate to the one paragraph. Each question has **only one** correct answer among the four given options (A), (B), (C) and (D).

Paragraph for Questions 09 & 11

Read the following write up carefully and answer the following questions:

A plane p contains the line L_1: $\frac{y}{b}+\frac{z}{c}=1$, x = 0 and is parallel to the line L_2: $\frac{x}{a}-\frac{z}{c}=1$, y = 0

9. Equation of plane p is

(A) $\frac{x}{a}-\frac{y}{b}+\frac{z}{c}+1=0$ (B) $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}-1=0$

(C) $\frac{x}{a}+\frac{y}{b}-\frac{z}{c}-1=0$ (D) $\frac{x}{a}-\frac{y}{b}-\frac{z}{c}+1=0$

10. If the shortest distance between L_1 and L_2 is $\frac{1}{4}$ then the value of $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}$ equals

(A) 16 (B) 64

(C) 128 (D) 192

11. Distance of image of A(a, 0, 0) in the plane p from $M\left(-\frac{5}{3},\frac{8}{3},\frac{11}{3}\right)$, where a = b = c = 1, is equal to

(A) 1 (B) 2

(C) 3 (D) 4

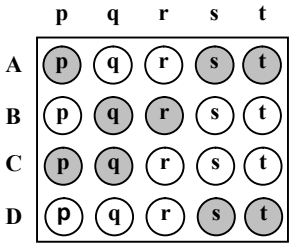
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SECTION - B

Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Match the following Column-I with Column-II

Column – I		Column – II	
(A)	f(x) = x[x] (where [.] denotes the greatest integer function and |.| denotes modulus function)	(p)	f(x) is continuous and differentiable everywhere on real domain
(B)	f(x) = x|x|	(q)	f(x) is continuous and differentiable at x = 0
(C)	f(x) = x + |x|	(r)	f(x) is continuous but not differentiable at x = 0
(D)	f(x) = x + [x]	(s)	f(x) is either discontinuous or non-differentiable for more than one $x \in R$
		(t)	f(x) is a periodic function

2 Match the following Column-I with Column-II

Column – I		Column – II	
(A)	Values of k for which the origin and the radical centre of three circles described on there sides 3x – 4y + 5 = 0, x + y + 5 = 0 and 4x + 3y – 10 = 0 of Δ as diameter lies opposite to the line 4x – ky + 1 = 0	(p)	2
(B)	Values of λ for which the pair of lines $x^2 + 2\lambda xy + 2y^2 = 0$ and $(1 + \lambda)x^2 – 8xy + y^2 = 0$ are equally inclined to each other	(q)	3
(C)	Find the points on the axis of parabola $y^2 – 3x + 6y + 15 = 0$ such that three distinct normals can be drawn	(r)	–2
(D)	Let A and P represents the area and perimeter of the triangle formed by the common tangents to the circles $x^2 + y^2 – 6x = 0$ and $x^2 + y^2 + 2x = 0$. Then value of $\frac{A}{\sqrt{3}}, \frac{P}{\sqrt{3}}$	(s)	6
		(t)	4

Space for rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Let $f(x) = x^{\alpha} + 3x^{\beta}$ with $\alpha > 1$, $\beta > 1$. The value of 'a' for which the area of the figure bounded by the graph of $y = f(x)$ and straight lines $x = 0$, $x = 1$ and $y = f(a)$ have the greatest value is A and the value for which area is minimum is B. Find the value of $A + 2B$ _____

2. Let $S = \frac{1^2}{1.3} + \frac{2^2}{3.5} + \frac{3^2}{5.7} + ... + \frac{500^2}{999.1001}$. If where [.] denotes the greatest integer function then the number of total divisors of [S] is _____

3. If $\int_0^{\pi} \left(\sin x - ax\left(\pi - x\right)\right)^2 dx$ is minimum for $a = \frac{120}{\alpha\pi^{\beta}}$ then the value of $\alpha + \beta$ is _____

4. Consider the polynomials $P(x) = x^6 - x^5 - x^3 - x^2 - x$ and $Q(x) = x^4 - x^3 - x^2 - 1$. Given that z_1, z_2, z_3 and z_4 are the roots of the equation $Q(x) = 0$. The value of $P(z_1) + P(z_2) + P(z_3) + P(z_4)$ is _____

5. Let $\vec{A} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{B} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{C} = \hat{j} + \hat{k}$. If the vector $\vec{B} \times \vec{C}$ can be expressed as a linear combination $\vec{B} \times \vec{C} = x\vec{A} + y\vec{B} + z\vec{C}$ where x, y, z are scalar, then find the value of $x + y + z$ _____

6. If w is the imaginary cube root of unity, then the number of ordered pairs of integers (a, b) such that $|aw + b| = 1$ is _____

Space for rough work